

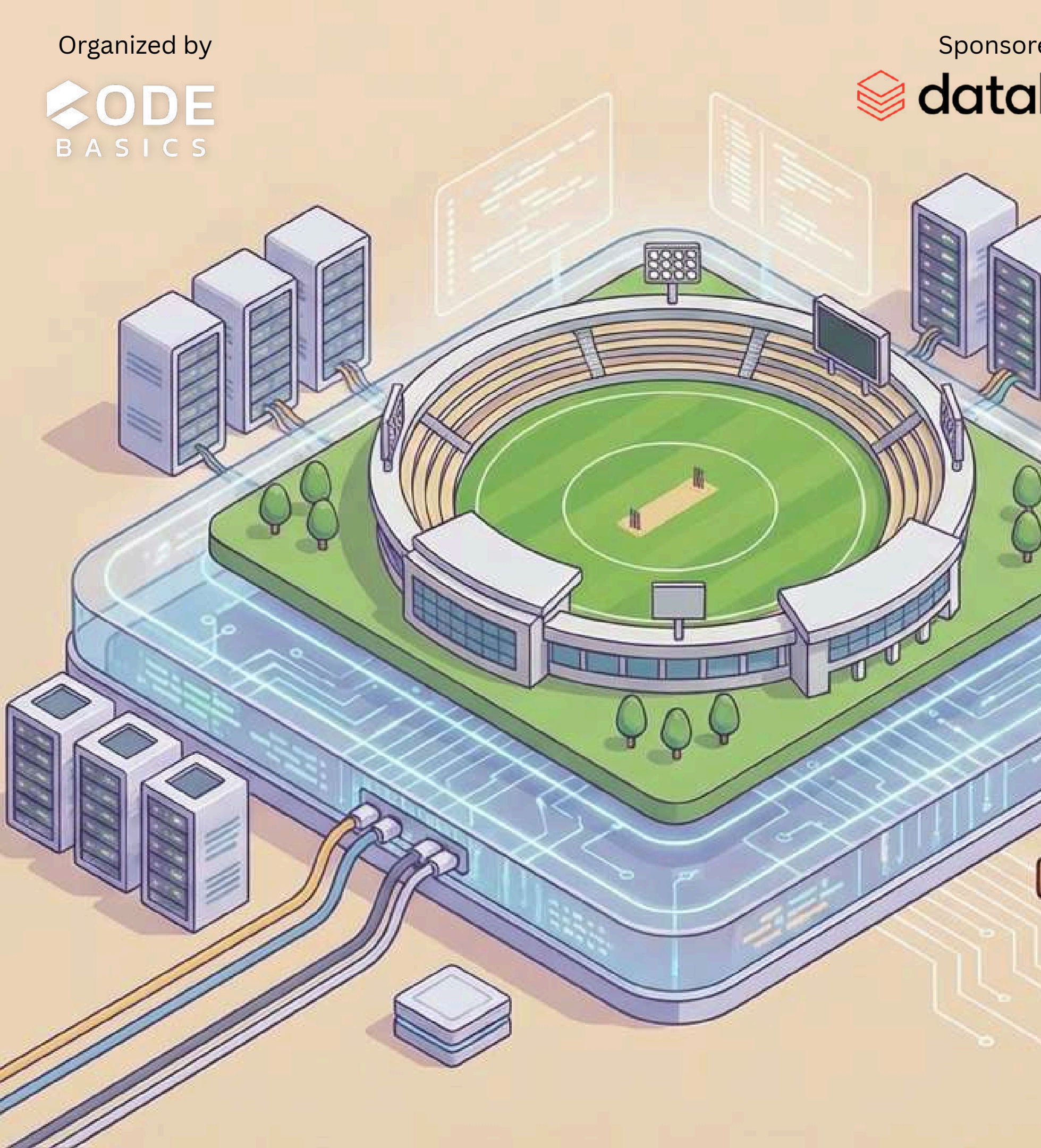
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An isometric illustration of a cricket stadium. The stadium is green with a brown pitch and is surrounded by tiered seating. It is connected to several server racks and data lines, suggesting a high-tech or data-driven environment. The stadium is set on a blue base with circuit-like patterns.

IPL Pulse: End-to-End Analytics & Prediction Engine

Presented by Divya Tiwari





ABOUT ME:

Divya Tiwari

Data Analyst/BI Analyst

-  Tech Stack:
 - Languages: Python (PySpark), SQL
 - Tools: Power BI, Excel, Databricks
 - Core: Data Warehousing, ETL, Data Viz
-  Connect:
 - LinkedIn- [linkedin.com/in/divyatiwari77](https://www.linkedin.com/in/divyatiwari77)
 - GitHub- [divyatiwari77](https://github.com/divyatiwari77)



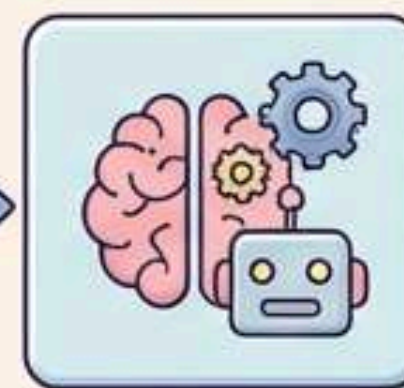
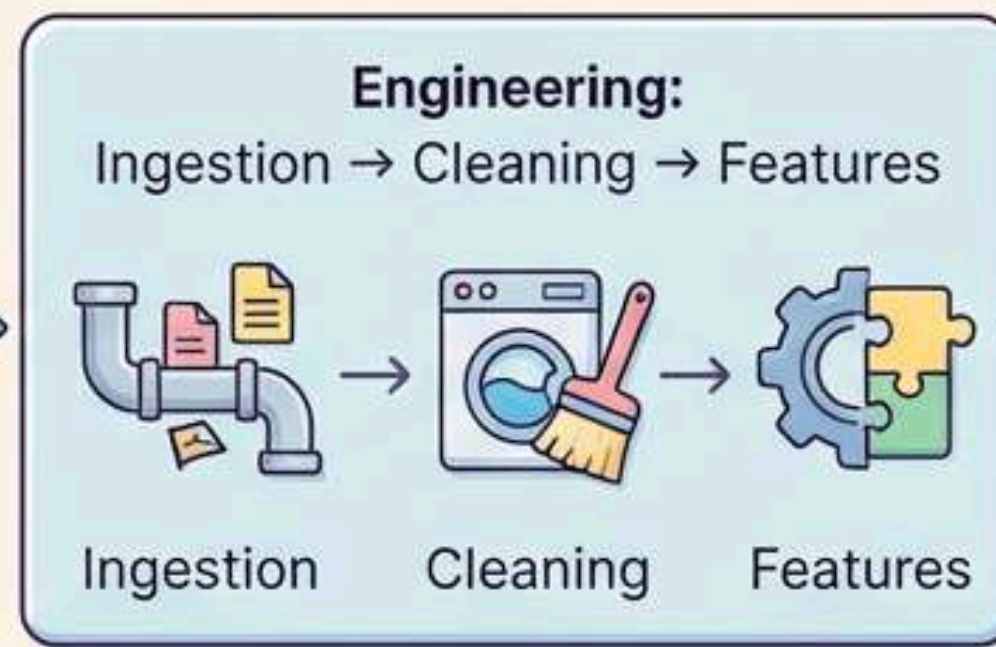
PROJECT CONTENTS:



The Problem :
Data Chaos &
Uncertainty



The Solution:
Lakehouse
Architecture



Intelligence:
ML Model
Training



Impact:
Insights &
Dashboard

 databricks



THE PROBLEM: DATA CHAOS

■ The Volume:



- 18 Years of History (2008-2025).



- Millions of ball-by-ball records.

■ The Issue:



- **Unstructured:** Scattered CSV files & Missing Values.



- **No Single Source of Truth:** Data is isolated.

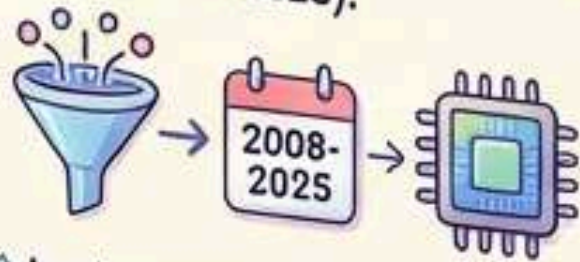


- **Intuition Over Data:** Decisions (Captaincy/Fantasy) are based on "Gut Feeling", not facts.

THE PROJECT OBJECTIVES

Build a Scalable Lakehouse

- ◆ Ingest & Process 18 years of IPL data (2008-2025).

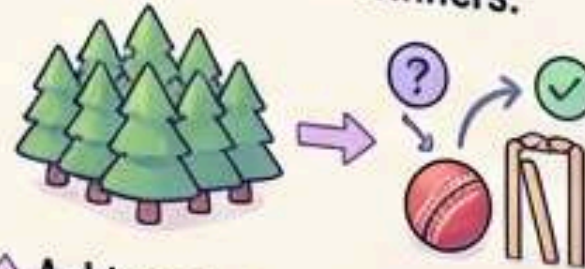


- ◆ Implement Medallion Architecture (Bronze → Silver → Gold).



Predict with AI

- ◆ Develop a Random Forest model to predict Match Winners.



- ◆ Achieve better accuracy than random probability (>50%).



Deliver Insights

- ◆ Identify high-impact players for Fantasy Leagues.



- ◆ Analyze Venue & Toss bias for Captains.



THE MEDALLION ARCHITECTURE



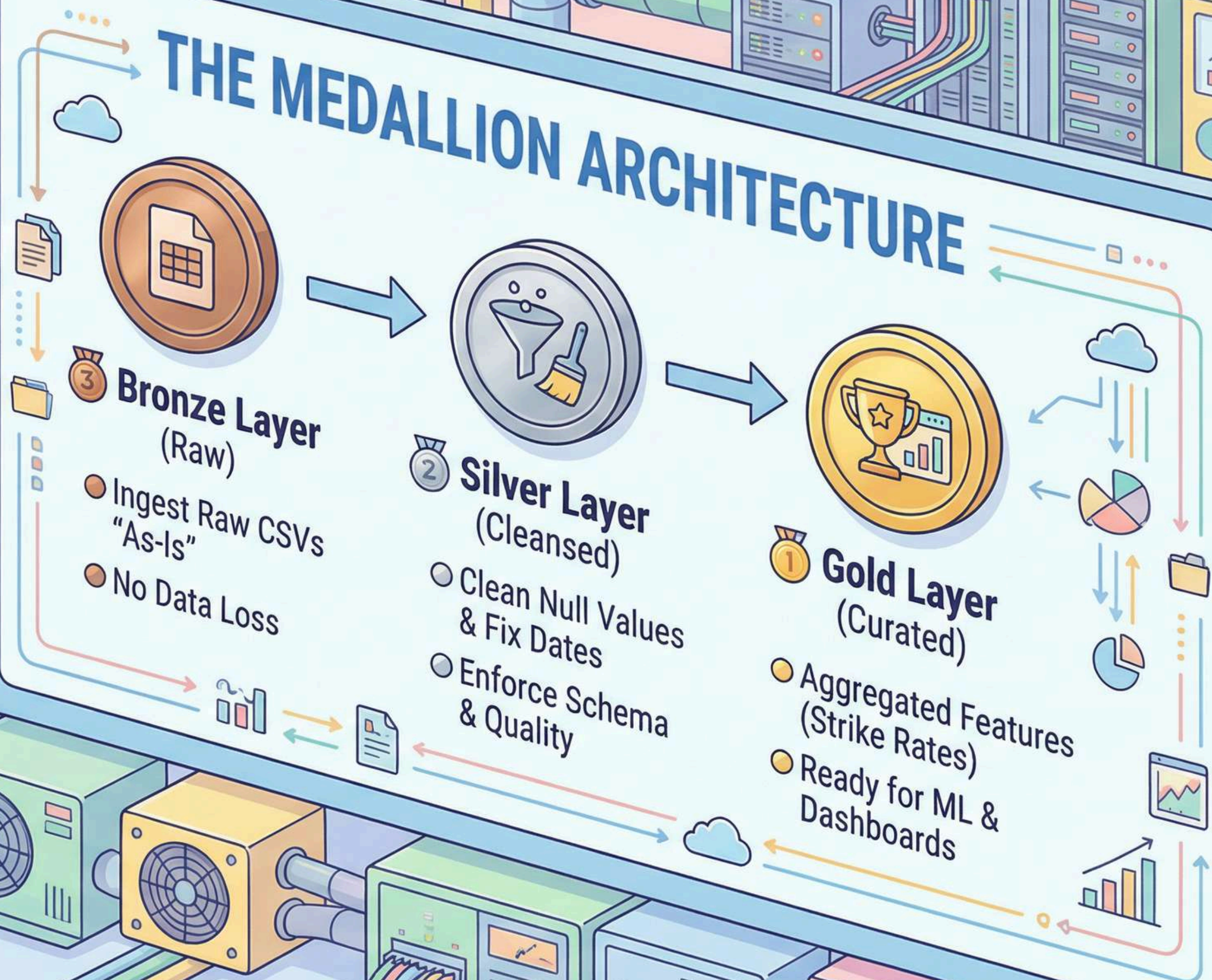
- Ingest Raw CSVs "As-Is"
- No Data Loss



- Clean Null Values & Fix Dates
- Enforce Schema & Quality



- Aggregated Features (Strike Rates)
- Ready for ML & Dashboards



UNITY CATALOG INGESTION

SECURE DATA ACCESS



Context:

- Using Unity Catalog Volumes for governance.



- Listing files to verify raw data availability.



```
# 1. Check Volume Path  
display(dbutils.fs.ls("/Volumes  
/workspace/default/ipl/"))
```


BRONZE LAYER: OUR LANDING ZONE



RAW DATA INGESTION

Context:

- Loading CSV files into Delta Tables.
- **Strategy:** "As-Is" Load (No modifications) to preserve history.



CSV FILES /
RAW SOURCE



DELTA TABLES
(BRONZE)



BRONZE
LAYER

```
$ _  
  
# Reading CSV & Saving as Bronze Table  
df_matches = spark.read.csv('/Volumes/.../matches.csv',  
header=True)  
df_matches.write.mode('overwrite').saveAsTable('bronze_matches')
```


SILVER LAYER: THE CLEANING ZONE



DATA TRANSFORMATION

Context:



- **Date Parsing:**
String → Date Format.

- **Null Handling:**
Replacing missing values in "City" & "Winner".

```
# Transformation: Date Format & Null Handling
df_matches_silver = df_matches \
    .withColumn("date", to_date(col("date"), "yyyy-
    'MM-dd")) \
    .withColumn(
        "city", when(col("city").isNull(),
        "Unknown").otherwise(col("city")))
    .withColumn("winner", when(col("winner").isNull(),
        "Unknown").otherwise(col("winner")))
```


SILVER LAYER: THE QUALITY CHECK



SCHEMA ENFORCEMENT



123

- **Type Casting:** Converting Runs/Wickets to Integers.



match_id

- **Renaming:** Standardizing id to match_id.

RAW
DATA
BLOCK

```
# Schema Enforcement: Casting to Integer  
df_deliveries_silver = df_deliveries \\\n    .withColumn("total_runs", \\\n        col("total_runs").cast("integer")) \\\n    .withColumn("is_wicket", col("is_wicket").\n        .cast("integer"))
```


GOLD LAYER: FEATURE ENGINEERING



Feature Engineering



Context:

- Creating Business Metrics.

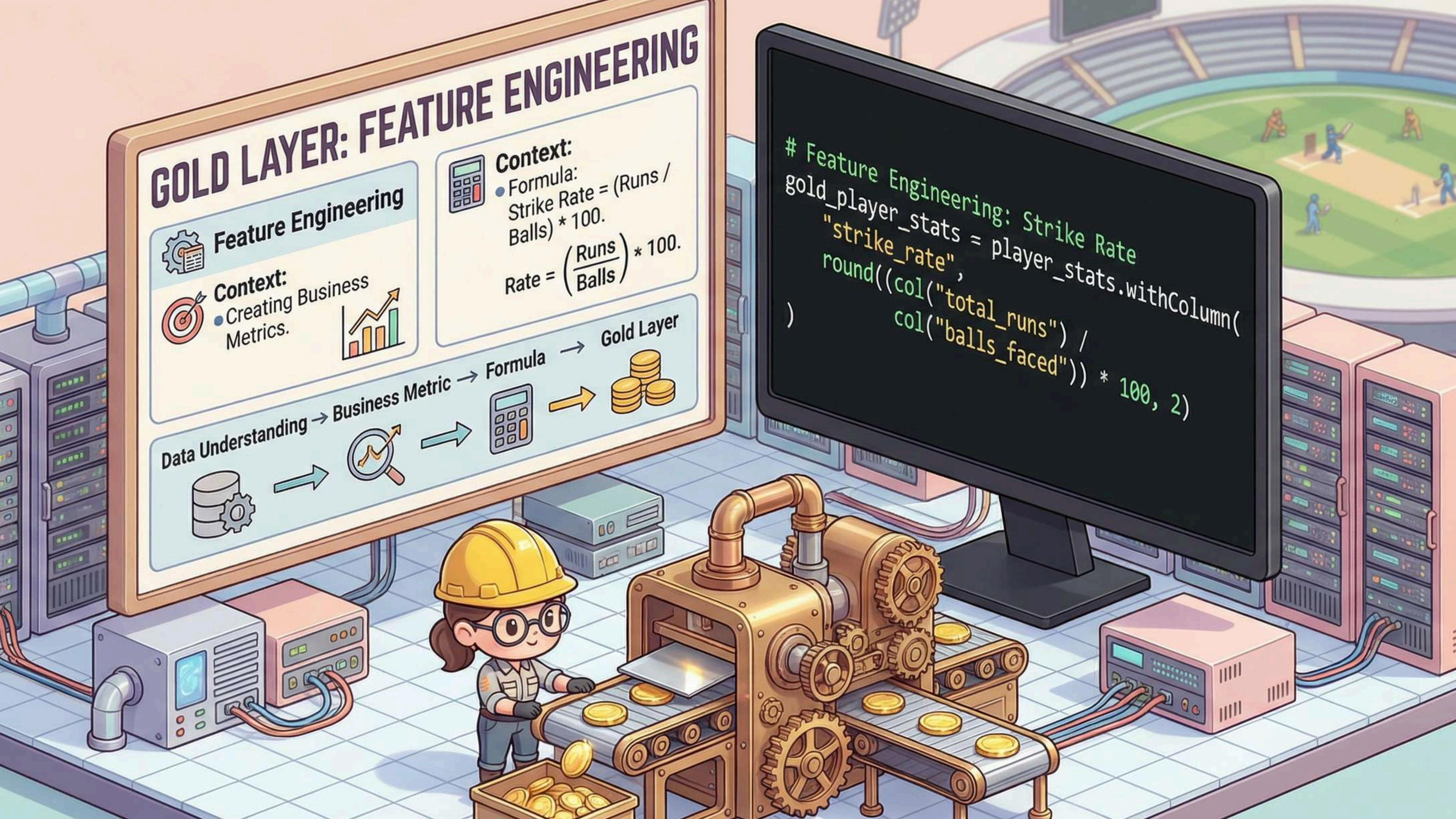
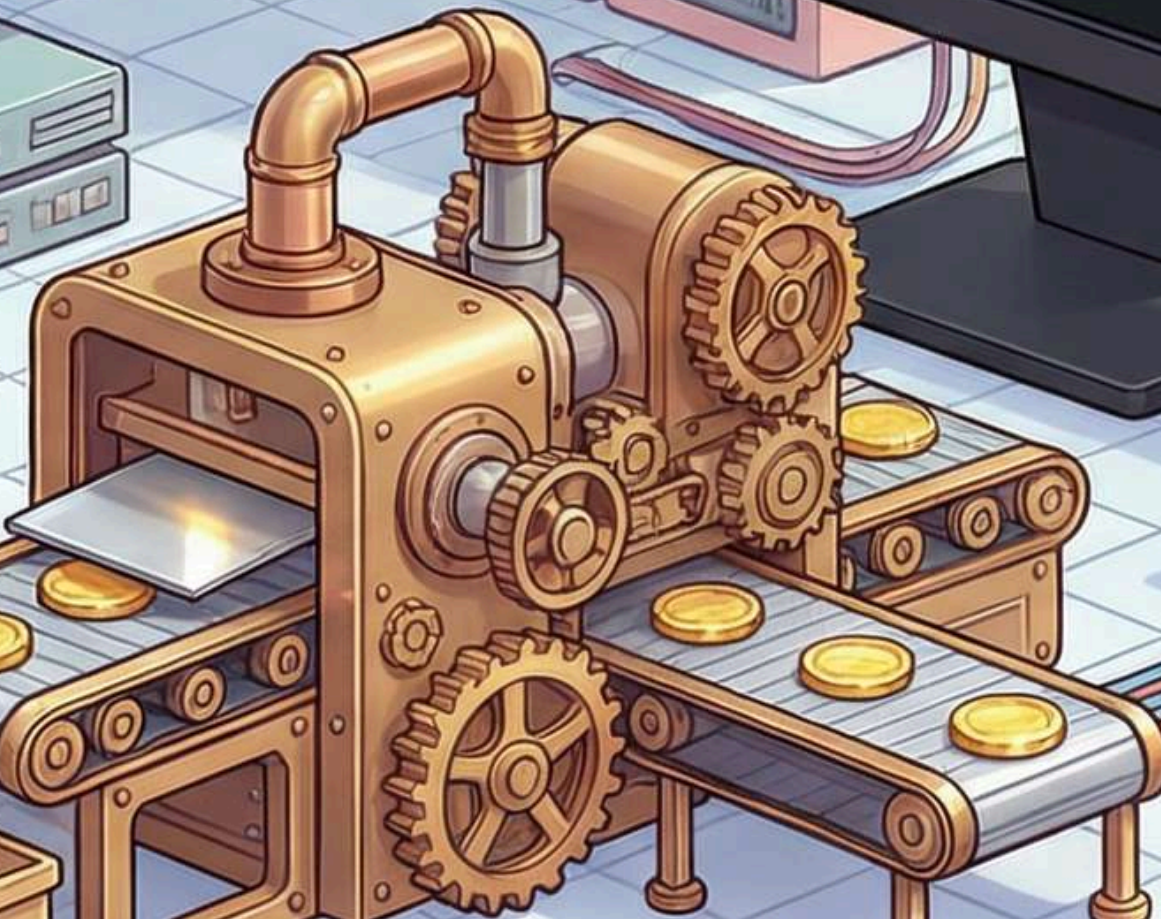


Context:

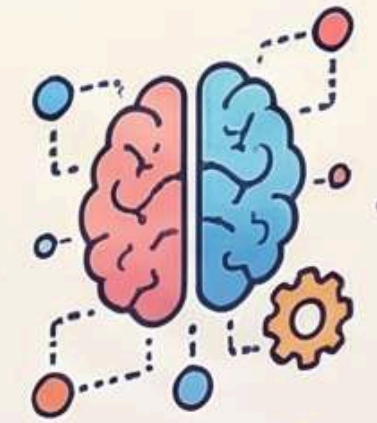
- Formula:
Strike Rate = $(\text{Runs} / \text{Balls}) * 100$.
- $$\text{Rate} = \left(\frac{\text{Runs}}{\text{Balls}} \right) * 100.$$



```
# Feature Engineering: Strike Rate
gold_player_stats = player_stats.withColumn(
    "strike_rate",
    round((col("total_runs") /
            col("balls_faced"))) * 100, 2)
)
```



GOLD LAYER: ML PREPERATION



ML Data
Preparation



Context

Selecting Predictors



Team



Venue



Toss

Filtering



Removing rows with Null values
for clean training.

```
# Selecting Features for ML Model  
ml_features_df = df_matches.select(  
    "team1", "team2", "venue",  
    "toss_decision", "winner",  
).dropna()  
>.
```

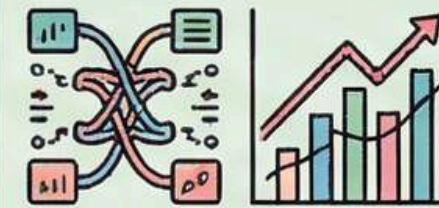


MODEL SELECTION: The Strategy

Why Random Forest?



● **Handles Categories:** Best for Team Names & Venues.



● **Non-Linear:** Captures complex game patterns.



● **Robust:** Less overfitting than Decision Trees.

MODEL TRAINING



Context:



• **Algorithm:**
Random Forest
Classifier.



• **Split:**
80% Training Data /
20% Testing Data.



Model
Training
Process

```
* Training the Model  
rf = RandomForestClassifier(labelCol="label",  
                             featuresCol="features")  
rf_model = rf.fit(train_data)
```


MLFLOW EVALUATION



Performance Metrics-



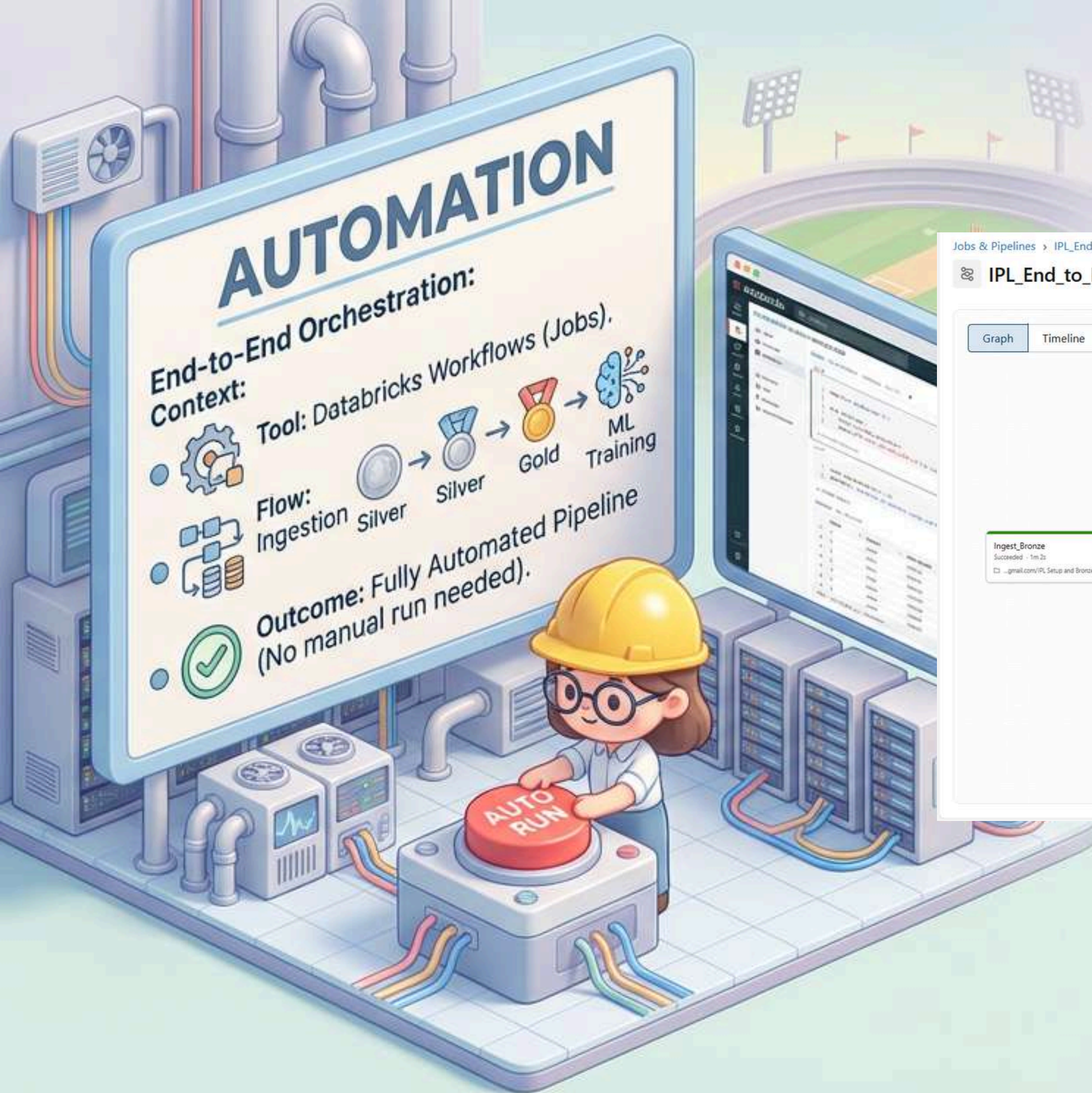
- **Metric:** ~55% Accuracy Score.



- **Tracking:** Logged via MLflow for reproducibility.

```
# MLflow Evaluation  
print(f"Accuracy: {accuracy*100: .2f}%")  
mlflow.log_metric("accuracy", accuracy)
```





Jobs & Pipelines > IPL_End_to_End_Workflow >

IPL_End_to_End_Workflow run [Send feedback](#)

Delete job run

Graph Timeline List



Job run details

Job ID [209104279962228](#)

Job run ID [1059269382196127](#)

Launched Manually

Started Jan 27, 2026, 11:26 AM

Ended Jan 27, 2026, 11:28 AM

Duration 2m 40s

Execution time 2m 37s

Queue duration -

Status ✓ Succeeded

Lineage No lineage information for this job
[Learn more](#)

Performance optimization Enabled

[View run events](#)

Compute

[Serverless](#)

[Logs](#)

INSIGHTS And VISUALIZATION

Strategic Insights-

Context:



- **Venue Bias:** Teams Chasing at Chinnaswamy Stadium have a 60% Win Rate.



- **Player Impact:** Identified high-value finishers (High Strike Rate) for Fantasy Picks.



CHINNASWAMY STADIUM
WIN RATE (CHASING)



HIGH-VALUE FINISHERS
(STRIKE RATE vs. IMPACT SCORE)



IPL Setup and Bronze layer × IPL Silver Cleaning IPL Gold Aggregation IPL_Machine_Learning_Model IPL Analytics 2026 +

File Edit View Run Help Python ▾ Tabs: ON ▾ ☆ Last edit was 3 days ago Run all Serverless ▾ Schedule (1



	A ^B _C path	A ^B _C name	i ² ₃ size	i ² ₃ modificationTime
1	dbfs:/Volumes/workspace/default/ipl/deliveries.csv	deliveries.csv	27019953	1769454628000
2	dbfs:/Volumes/workspace/default/ipl/matches.csv	matches.csv	225266	1769454615000

↓ ▾

 2 rows | 32.82s runtime

Refreshed 3 hours ago

✓ 3 days ago (9s)

2

```
# 2. Database (Schema) set
spark.sql("CREATE SCHEMA IF NOT EXISTS workspace.default")
spark.sql("USE workspace.default")

# 3. Reading Matches File (Volume Path)
df_matches = spark.read.csv("/Volumes/workspace/default/ipl/matches.csv", header=True, inferSchema=True)
df_matches.write.mode("overwrite").saveAsTable("bronze_matches")

# 4. Reading Deliveries File
df_deliveries = spark.read.csv("/Volumes/workspace/default/ipl/deliveries.csv", header=True, inferSchema=True)
df_deliveries.write.mode("overwrite").saveAsTable("bronze_deliveries")

print("Tables are ready in Unity Catalog!")
```

> See performance (4)

> df_matches: pyspark.sql.connect.dataframe.DataFrame = [id: integer, season: string ... 18 more fields]

+ New

Home

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Jobs & Pipelines > IPL_End_to_End_Workflow >

IPL_End_to_End_Workflow run [Send feedback](#)

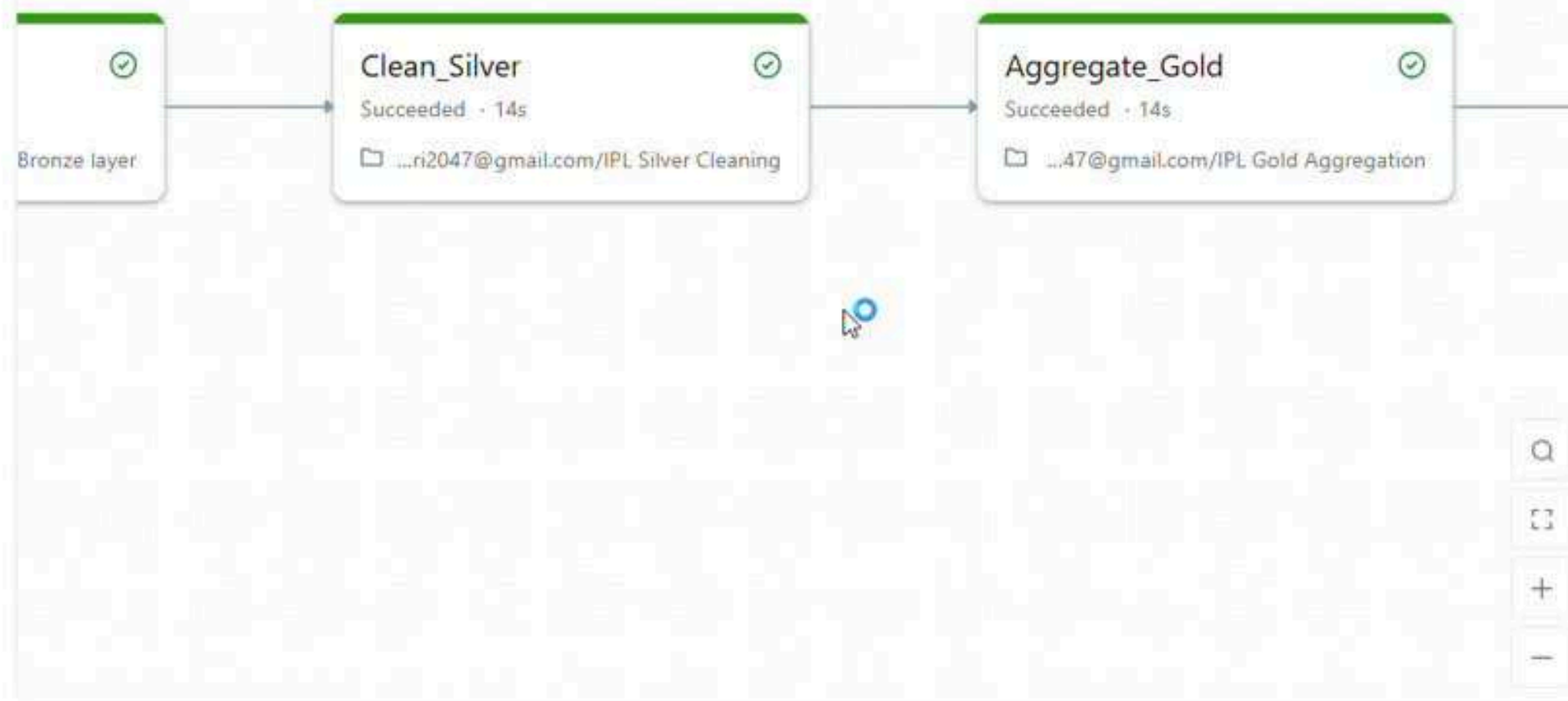
Delete job run

Repair run

Graph

Timeline

List



Job run details

Job ID [209104279962228](#)

Job run ID [812193476840144](#)

Launched **Manually**

Started **Feb 01, 2026, 07:21 PM**

Ended **Feb 01, 2026, 07:23 PM**

Duration ⓘ **2m 6s**

Execution time ⓘ **1m 35s**

Queue duration ⓘ **28s**

Status **✓ Succeeded**

Lineage ⓘ **5 upstream tables, 6 downstream tables**

Performance optimization ⓘ **Enabled**

View run events

Compute

Serverless

Logs

BUSINESS IMPACT AND VALUE

IMPACT ON IPL/CRICKET WORLD



Captains:
Data-backed
Toss Decisions



Fantasy Players:
Pick High-Impact
Players



Analysts:
Automated
Pre-match Reports





FUTURE PROSPECTUS OF THIS PROJECT



REAL-TIME ANALYTICS:

Integrating Spark Structured Streaming for instant, ball-by-ball win probability calculations.



GENAI INTEGRATION:

Developing a "Chat with Data" bot to answer natural language queries like, 'How does Kohli perform against Bumrah?'.



DEPLOYMENT:

Exposing the predictive model via robust REST APIs for seamless mobile app integration.

Thank You!

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BUILD WITH DATABRICKS

Hands-On Project Challenge

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